MIL-C-51232C(EA)
7 February 1977
SUPERSEDING
MIL-C-51232B(MU)
19 March 1971

MILITARY SPECIFICATION

CLOTH, MARQUISETTE, POLYESTER

This specification is approved for use by Edgewood Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of a woven polyester cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied-Yarns (Army and Numbered).

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections.
FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-191 - Textile Test Methods.

FSC 8305

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Edgewood Arsenal, Attn: SAREA-TS-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Fibers. The fibers shall be high quality polyester staple containing black pigment. The fiber size shall be 2.25 denier per filament and the staple length shall be 1.5 inches.

3.1.2 Yarn. The staple fiber shall be spun into uniform yarn, free of fuzz balls, using a cotton or other suitable yarn spinning system. The warp yarn shall be seven singles (7/1) count and the filling yarn shall be four singles (4/1) count. Both yarns shall have a 4.0 twist multiplier, either direction.

3.2 Construction.

3.2.1 Weave. The yarns shall be uniformly woven into a marquisette fabric. The weave pattern shall be a two end leno, repeating on two picks, as shown in figure 1.

3.2.2 Yarns per inch. The finished cloth shall have 36 $\pm$ 2 warp ends per inch and 15 $\pm$ 1 pick per inch in the filling when tested as specified in 4.4.4.

3.2.3 Selvages. The selvages, if used, shall be at least 1/4 inch but no greater that 1/2 inch in width. They shall be woven of the same fiber as the body of the fabric.

3.2.4 Width. The width of the cloth, including selvages, shall be as specified by the procuring agency, but in no case less than 36 inches (see 6.2).

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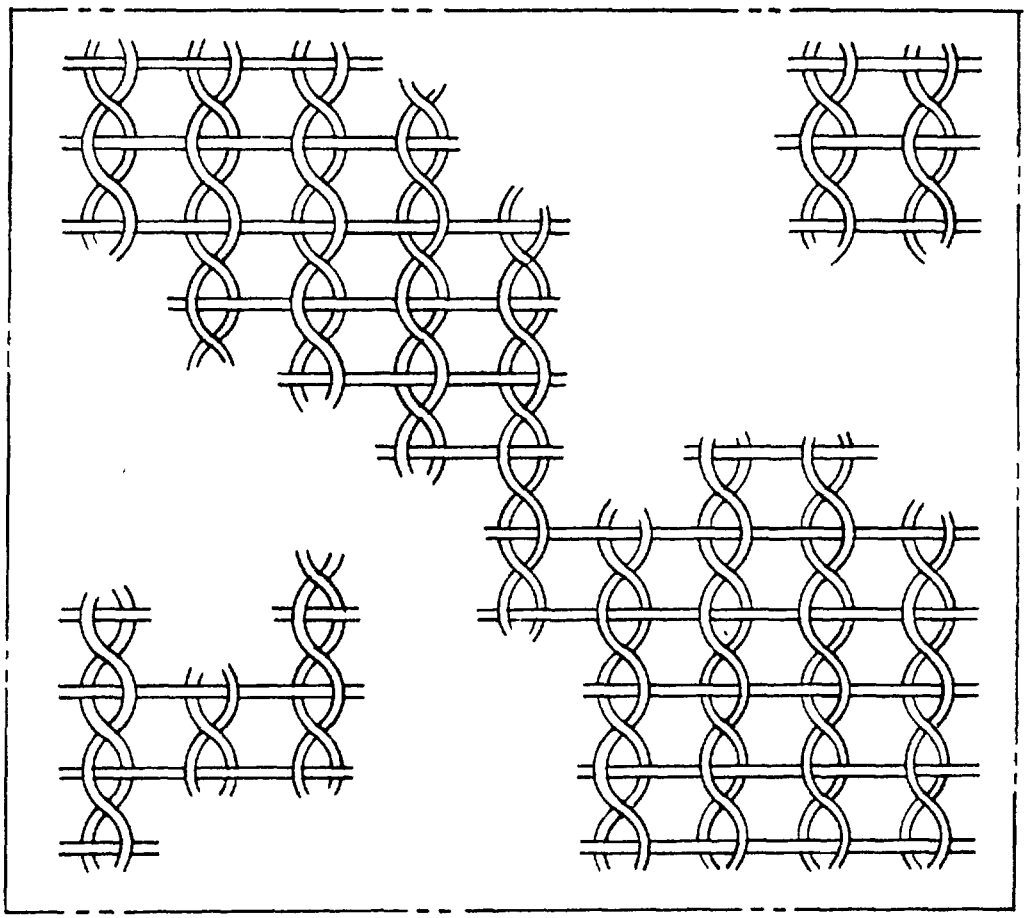


Figure 1. - Leno Weave

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3.3 Finishing.

3.3.1 Scouring. The cloth shall be scoured to remove all sizing materials.

3.3.2 Heat set. The cloth shall be heat set so as to achieve maximum stability of the weave.

3.3.3 Napping and shearing. The cloth shall be napped and sheared on both sides to achieve a uniform fibrous surface (see 6.3). The degree of napping shall be as shown in figure 2.

3.3.4 Water repellency. The cloth shall be treated with a durable water repellent finish.

3.4 Physical properties.

3.4.1 Weight. The weight of the finished cloth shall be 6.5 (+ 1.0 or -0.5) ounce per square yard when tested as specified 4.4.4.

3.4.2 Bursting strength. The bursting strength of the finished cloth shall be a minimum of 70 pounds per square inch when tested as specified in 4.4.4.

3.4.3 Yarn slippage. After testing as specified in 4.4.4 the mesh openings between the warp yarns shall not exceed 0.4 inches and those between the filling yarns shall not exceed 0.06 inches.

3.4.4 Air permeability. The air permeability of the finished cloth shall be greater than 500 cubic feet per minute per square foot when tested as specified in 4.4.4.

3.4.5 Abrasion resistance. The thickness of the cloth shall not decrease more than 10 percent when tested as specified in 4.4.4.

3.4.6 Water resistance. The cloth shall have a rating of 70 or higher when tested as specified in 4.4.4.

3.4.7 Burning rate. The rate of burning of the complete specimen shall be less than 15 inches per minute in both the warp and the filling direction of the cloth when tested as specified in 4.4.4.

3.4.8 Mildew resistance. The cloth shall show no growth when tested as specified in 4.4.4.

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Figure 2. - Polyester Marquisette Cloth (Magnified 18 times)

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3.5 Size. Unless otherwise specified, the cloth shall be furnished in rolls of 100 ± 50 yards (see 6.2).

3.6 Proproduction. Prior to the start of regular production, a preproduction sample of cloth shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2.1, 3.2.3, 3.2.4, 3.3, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 50 yards of cloth shall be manufactured using the same methods, materials, equipment, and processes as will by used in regular production.

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4.3.2 Inspection procedure for examination and test. The preproduction sample shall be examined and tested for all the requirement of this specification. A 1-1/2-yard specimen, full width, cut from each 10-yards of the sample shall be used for testing as specified in 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet all the requirements of this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to the start of regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the cloth produced by one manufacturer, at one plant, from the same materials, using the same processes, and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For yard-by-yard examination. Sampling for total yardage shall be conducted in accordance with MIL-STD-105. The unit of product shall be 1 linear yard. The number of rolls from which the sample is to be taken shall be in accordance with table I.

TABLE I. Sample rolls

Lot size in yards	Sample size in rolls
Up to 1301	3
1301 to 3,200	5
3,201 to 8,000	7
8,001 to 22,000	10
22,001 to 110,000	15
110,001 to 530,000	25
530,001 and over	50

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-4. The unit of product shall be one roll. A strip 1-1/2 yards in length and full width shall be taken from each sample roll for testing.

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4.4.3 Inspection procedure.

4.4.3.1 For yard-by-yard examination. The sample cloth shall be examined in accordance with table II and MIL-STD-105 on the basis of defects per hundred units. An approximate equal number of yards shall be examined from each roll selected. The defects listed in table II shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs.

4.4.3.2 For destructive tests. Cloth samples shall be tested in accordance with table III; the acceptance number is zero.

4.4.4 Tests. The methods of testing specified in FED-STD-191, listed in table III, shall be followed.

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TABLE II. Defects

Glossary No., FED-STD-4; Section III	Defects	Classification	
		Major AQL 6.5	Minor AQL 10
1, 2, 4, 26	Coarse filling bar or thick places in the warp or filling directions clearly visible* and exceeding 2-inches in length	X	
3, 5, 6, 7, 8, 27, 28, 29, 75	Slough off, jerked in filling, slubs, gouts, knots, woven-in waste, or flyer		X
9, 10, 31, 32	Light filling bar or light places in the warp or filling directions clearly visible* and exceeding 2-inches in length	X	
11, 12, 13, 16, 17, 18, 30, 33, 36, 37	Broken, missing, loose, or tight yarns:		
	Three or more contiguous, 2-inches or more in combined warp and filling directions	X	
	Two contiguous, less than 2 inches in combined warp and filling directions or singles, any length		X
14, 34	Floats, skips, or harness drops:		
	Multiple, 2-inches or more in combined warp and filling directions	X	
	Multiple, less than 2 inches in combined warp and filling direction or singles 2 inches or more in length		X
76	Smash or breakout	X	
77	Abrasion mark, bruise, chafe, or rub	X	
82, 83, 84	Hole, cut, tear, or break	X	

*Clearly visible at normal inspection distance (approximately three feet).

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TABLE III. Test methods

Characteristic	Requirement reference	Test method
Yarns per inch*	3.2.2	5050
Weight	3.4.1	5041
Bursting strength	3.4.2	5120
Yarn slippage	3.4.3	5410
Air permeability	3.4.4	5450
Abrasion resistance**	3.4.5	5302
Water resistance	3.4.6	5526
Burning rate	3.4.7	5906
Mildew resistance	3.4.8	5750

*At least 3 inches of yarn shall be counted for each determination.

**Specimen shall be subjected to a minimum of 2000 strokes of multi-directional abrasion. Abradant shall be cotton duck cloth conforming to type I, hard texture, No. 10, or type III, weight 12.29 ounces per square yard, of CCC-C-419. Abrading surface of abradant shall be moved after 2000 strokes. The area of cloth to be abraded will be measured before and after abrasion with a dial-type indicator having a 0.25 inch diameter presser foot and a three ounce load. The results of five measurements shall be reported to the nearest 0.001 inch. Pills of matted fibers interfering with proper measurement after abrading may be removed.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking, interplant shipment (see 6.5). Rolls of cloth shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123.

6. NOTES

6.1 Intended use. The cloth is intended for use in the inlet valve assembly of the ABC-M17 and M17A1 masks to prevent the passage of ice particles into the masks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Cloth width (see 3.2.4).

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(c) Length of roll if other than specified (see 3.5).
(d) Preproduction.

(1) Time allowed for supplier submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Standard sample. A standard sample can be used for comparing the napping and shearing and will be furnished by the Government. If required, the standard sample should be obtained as directed by the contracting officer.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 21, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

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Preparing activity:

Army - EA

Project No. 8305-A592

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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